The scenario: A general hospital system

The workflow follows a patient's admission to the hospital, through either emergency or general admission, all the way to initial treatment.

Environment: The hospital receives patients in two main ways: emergency admission, where paramedics bring the patient in, or general admission, where the patient walks in and is registered by the secretary. This scenario covers both paths. The emergency path starts when the paramedics pick up the patient. The general path starts when the patient arrives at reception. The two paths have different early stages but then join. Once a patient has been taken in by the nursing staff, both paths share the same doctor acceptance, assessment, room allocation and cleaning stages, until the patient is assessed in a room, ICU or surgery, or is discharged.

Participants and roles: The main participants focused on include:

- **Paramedics** (emergency only): pick up the patient, transport them and update the transport status.
- **Secretary:** registers incoming emergency patients and walk-in patients, and notifies the nursing staff.
- **Nursing staff:** triage walk-in patients, receive emergency patients, prepare patients, and coordinate and communicate with the doctor.
- **Doctor:** accepts the cases, assesses the patient and decides on the room.
- **Cleaning staff:** clean the area after the patient leaves.

Major pieces of work:

Emergency admission path

1. Paramedic pickup and transportation
2. Patient registration on arrival
3. Room or bay preparation (in parallel with transportation)
4. Patient handover from paramedic to nurse
5. Nurse intake and preparation

General admission path

1. Patient arrival and registration at reception
2. Nurse triage (vital signs taken and priority level assigned)
3. Waiting queue, ordered by triage priority
4. Nurse intake and preparation

Shared stages (both paths)

1. Doctor case acceptance
2. Doctor assessment and treatment
3. Room allocation by the doctor (admit to a general room, ICU or surgery, or discharge)
4. Room departure cleaning

Decisions and routing points: Deciding on the admission route (emergency or general). Assigning a triage priority level to walk-in patients. Whether a walk-in patient's condition is critical enough to be moved onto the emergency path. Picking the room type the patient should be routed to. Whether the doctor accepts the case or escalates the patient to a specialist. Whether the patient is discharged, admitted, or transferred to ICU or surgery.

Approval:

- **Hospital admission approval:** the hospital formally accepts the patient (both paths).
- **Doctor case acceptance:** needed before treatment begins. If it isn't accepted within a deadline, the case is passed on to another doctor. For walk-in patients, the deadline depends on the triage priority, so more urgent patients have shorter deadlines.
- **Room allocation approval:** ICU and surgery transfers require approval before the patient is moved.

Dependencies:

- Cleaning only begins after the patient is no longer in that area.
- Nurse intake can't begin until handover (emergency) or triage (general) is complete.
- Triage can't begin until the walk-in patient has been registered.
- Walk-in patients are seen in order of triage priority, not arrival time.
- Treatment can't begin until the doctor has accepted the case.

Failure and exception paths:

- **Paramedic transport fails or is delayed:** the hospital is informed, and preparation is paused or cancelled.
- **Triage finds a critical condition:** the walk-in patient is moved straight onto the emergency path.
- **A walk-in patient waits longer than their priority allows:** their priority is escalated.
- **A walk-in patient leaves before being seen:** the workflow ends in a "left without being seen" outcome.
- **The doctor doesn't respond within the deadline:** the case is rerouted to another doctor.
- **The patient's health deteriorates during intake:** escalated to emergency treatment.
- **The patient doesn't survive:** the workflow ends in a deceased-patient outcome.

External systems:

- **Ambulance dispatch system** (emergency path): paramedic status updates (picked up, en route, arrived).
- **Hospital patient records** (both paths): the patient's history is retrieved on arrival or registration and updated during treatment.

Expected outcome:

- The patient is admitted to a ward, transferred to ICU or surgery, discharged, recorded as deceased, or, for walk-ins only, recorded as having left without being seen. The full history is recorded and archived for future reference.
